Himank Arora

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Summary

Software engineer with 3+ years building backend systems, RESTful APIs, and data pipelines using Python, Java, and JavaScript, experienced in scalable architectures, database optimization, workflow automation, and AI-enabled applications.

Education

Northeastern University, Boston, MA May 2026
Master of Science in Information Systems
Relevant Coursework: Program Structures & Algorithms, Software Engineering, Data Science & ML, Database Design

Guru Gobind Singh Indraprastha University, New Delhi, India September 2021
Bachelor of Commerce (Honors)
Relevant Coursework: Information Systems Management, Computer Applications, Business Statistics, Business Mathematics

Technical & Professional Skills

Programming Languages: Python, Java, JavaScript, TypeScript, SQL, R, HTML, CSS
Frameworks & APIs: React, Redux, Next.js, Node.js, Express, Flask, FastAPI, Spring, RESTful APIs, GraphQL, Tailwind CSS, Clerk, Zod, WebRTC, Bootstrap, Material-UI
Databases & Data Technologies: MySQL, PostgreSQL, PostGIS, Supabase, MongoDB, Oracle, SQLite, Redis
AI / Machine Learning & GenAI: LLM APIs (OpenAI, Google Gemini AI), Prompt Engineering (Zero-shot, Few-shot, Chain-of-Thought), Feature Selection (LASSO, PCA), Model Evaluation & Benchmarking
Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Scikit-learn, XGBoost, Tableau, Power BI, Jupyter, MS Excel, Google Analytics
Cloud & DevOps: AWS, Vercel, Docker, GitHub Actions, CI/CD Pipelines
Tools & Platforms: Git, GitHub, Jira, Postman, VS Code, Anaconda, Figma, WordPress
Soft Skills: Problem Solving, Analytical Thinking, Communication, Strategic Planning, Adaptability, Teamwork
Certificates: Python Data Structures, Git/GitHub, CS50 by Harvard University, Data Analysis with R Programming

Professional Experience

Ferro Star, New Delhi, India
Software Developer April 2022 – April 2024

- Engineered FastAPI backend for inventory management, processing 1,000+ monthly transactions with audit logging
- Redesigned MySQL schema to 3NF with B-tree indexes and connection pooling, cutting query response time by 60%
- Constructed Python/Pandas ETL pipeline normalizing supplier catalogs, eliminating 12+ hours of weekly data entry
- Implemented JWT auth and RBAC across 3 departments, securing inventory and pricing data with stateless API design
- Delivered React dashboard adopted by 4 departments and leadership for pricing and procurement decisions

Super Enterprises, New Delhi, India
Software Developer April 2021 – March 2022

- Launched React/Node.js website reaching 8,000+ monthly sessions (Google Analytics), improving engagement by 35%
- Designed RESTful API with MySQL backend for product catalog and pricing, cutting latency by 50% across 300+ SKUs
- Automated pricing and inventory sync via Python scripts, cutting update time from 90 to under 10 minutes
- Integrated third-party email notification APIs, reducing customer response time from 24 hours to under 2
- Developed React components with dynamic filtering and validation, reducing incomplete submissions by 28%

Projects

MatchRide | *Next.js 14, TypeScript, PostgreSQL, PostGIS, Supabase, Clerk, Google Gemini AI, WebRTC, Tailwind CSS, Vercel*

- Built and deployed full-stack carpooling platform serving 50+ users across 30+ endpoints
- Cut matching to sub-second using PostGIS LineStrings with GIST indexes over full-table distance scans
- Improved group match fairness with a home-proximity algorithm and 5-rating threshold preventing new-driver bias
- Achieved zero dropped real-time updates via Supabase Realtime plus WebRTC peer-to-peer voice signaling
- Eliminated hard AI search failures by chaining Gemini 2.5/2.0/1.5 Flash with Zod validation and fallback
- Secured deployment via Svix webhook verification, CRON_SECRET endpoints, and restricted API keys

MatchWise | *Python, FastAPI, React, MongoDB, LLM APIs, GitHub Actions*

- Architected semantic job matching platform achieving 45% higher resume-to-JD relevance over keyword search
- Designed 3-tier system (React, FastAPI, MongoDB) with JWT auth and RBAC, enforcing isolated data access per role
- Optimized async resume analysis pipeline in FastAPI delivering results in under 3 seconds under concurrent load
- Established CI/CD with GitHub Actions running 40+ tests on every push, catching regressions before merge

Train Reservation System | *SQL, PL/SQL, Database Design*

- Structured Oracle schema in 3NF with 12+ tables and foreign key constraints, eliminating 40% data redundancy
- Wrote 15+ PL/SQL packages automating booking validation, seat allocation, waitlist promotion, and cancellations
- Accelerated join performance with B-tree indexes, reducing execution from 2.8s to 0.4s – 86% improvement
- Enforced 4-tier RBAC and 10+ views delivering occupancy rates, revenue summaries, and performance metrics

FitHub AI | *Python, OpenAI API, Prompt Engineering, Streamlit, Jupyter*

- Engineered GPT-4 fitness assistant applying zero-shot and CoT prompting, improving output accuracy by 65%
- Applied structured JSON output schemas, eliminating free-text responses and ensuring consistent rendering
- Created evaluation pipeline testing prompt variations, measuring accuracy and output quality
- Shipped Streamlit interface delivering real-time AI-generated recommendations in under 4 seconds

Personal Dual Portfolio Website | *HTML, CSS, JavaScript, React*

- Built performance-optimized portfolio in React, achieving 96/100 Lighthouse score and sub-1.5s load time

Inventory Management System | *Java, JavaFX, MySQL, JDBC*

- Assembled desktop app managing 150+ products across procurement and sales, cutting reconciliation time by 50%
- Set up JDBC connection pooling with prepared statements, improving DB response by 45% and blocking SQL injection
- Implemented 3-tier RBAC with Stack-based undo/redo and automated invoice generation with dynamic tax calculations

Disentangled Representation Learning (InfoGAN & CGN) | *Python, PyTorch, GANs, TensorBoard, Jupyter*

- Engineered 4 InfoGAN variants (baseline, orthogonal, contrastive, combined), achieving 1.0 categorical accuracy
- Combined regularization achieved highest continuous code correlation at 0.913, up from 0.904 baseline
- Built CGN on Colored MNIST, disentangling shape, foreground, and background as independent causal factors
- Created custom metrics (mutual information, traversal linearity, factor independence) benchmarking all 4 variants

Medulloblastoma Classification Using Machine Learning | *Python, Scikit-Learn, Jupyter*

- Trained classifier achieving 93% accuracy on a high-dimensional 54,675-feature dataset across 4 cancer subtypes
- Reduced feature space 99.7% (54,675 to 150) via LASSO, PCA, and Random Forest, accelerating training by 82%
- Benchmarked SVM, XGBoost, Random Forest, and Logistic Regression via GridSearchCV, achieving 0.94 F1-score
- Surfaced 20 top biomarker genes per subtype via feature importance, visualized with heatmaps and PCA biplots

Mental Health Wellness Website | *React, Node.js, Express.js, MongoDB*

- Launched full-stack MERN platform with scheduling, resource library, and progress tracking across 3 roles
- Deployed Express REST API with 15+ endpoints, JWT auth, bcrypt password hashing, and RBAC middleware
- Streamlined booking UI in React/Material-UI, cutting appointment flow from 5 steps to 2 and time by 60%

Health Management System | *Java, Swing, MySQL*

- Set up healthcare supply chain for 85+ medical supplies across 5 departments, replacing manual tracking
- Applied 4-tier role-based access with input validation, eliminating invalid data entry at database level
- Created Swing interface with automated low-stock alerts triggering at 15% threshold, enabling reordering
- Structured normalized MySQL schema across 8 tables covering suppliers, stock levels, orders, and distribution

E-commerce Sales Data Analysis | *Python, Jupyter, Pandas, Matplotlib, NumPy*

- Processed 10,000-record dataset, resolving 480+ missing values and applying IQR outlier treatment for analysis
- Uncovered revenue drivers: top 3 months drove 47% of sales; 5 cities drove 58%, shaping inventory strategy
- Pricing elasticity analysis: 18% price reduction drove 32% volume increase, informing markdown strategy
- Produced 6 dashboards (time-series, heatmaps, correlation matrices) translating findings for stakeholder review

Do All | *Python, Flask, SQL, HTML, CSS, JavaScript*

- Developed productivity app unifying task management, notes, and calendar with multi-user auth and bcrypt
- Reduced frontend latency by 75% replacing full-page reloads with AJAX async updates via JavaScript/jQuery

Stopwatch App | *HTML, CSS, JavaScript, Bootstrap, jQuery*

- Created precision stopwatch using JavaScript timing APIs with start/stop/lap/reset and Bootstrap UI